

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO5: *Implement database operations using query processing, optimization, and transaction management to ensure consistency and reliability.* (BL3, BL4)

Activity Tool : Mock Interview (Low)
Assessment Tool Weightage: 30%

Dr

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Interviewer: 'Explain the steps of query processing in DBMS. How does a query optimizer decide the best execution plan?'	BL3 – Apply	5
2	Interviewer: 'What is the difference between heuristic optimization and cost-based optimization? When would you use each?'	BL4 – Analyze	5
3	Interviewer: 'How would you implement database connectivity in a real project? Explain the difference between JDBC and ODBC.'	BL3 – Apply	5
4	Interviewer: 'What are ACID properties? Give a real-world example where each property is critical.'	BL3 – Apply	5
5	Interviewer: 'Explain concurrency control problems: dirty read, lost update, and unrepeatable read. How does 2PL solve them?'	BL4 – Analyze	5
6	Interviewer: 'What is deadlock in DBMS? Describe the deadlock detection algorithm and at least two prevention strategies.'	BL4 – Analyze	5
7	Interviewer: 'Compare Immediate Update and Deferred Update recovery. In what scenarios is each preferred?'	BL4 – Analyze	5
8	Interviewer: 'Explain Shadow Paging. What are its advantages and limitations compared to log-based recovery?'	BL3 – Apply	5

9	Interviewer: 'What is a serializability schedule? How do you test for conflict serializability using a precedence graph?'	BL4 – Analyze	5
10	Interviewer: 'Describe the Two-Phase Locking protocol variations: Basic 2PL, Strict 2PL, and Rigorous 2PL. Which is most commonly used and why?'	BL4 – Analyze	5

Code of Conduct:

I Vys h n a v i . G (1 9 2 5 2 5 2 6 0) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Vyshnavi G

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively

Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

4. Acid Properties

Interviewer: Good morning. Please introduce yourself.

Candidate: Good morning, sir/mam. My name is Ashmi P S. I am currently pursuing my B.E. in Computer Science and Engineering. I am interested in software development and database technologies. I am also interested in learning new technologies and improving my technical skills.

Interviewer: That's good, Ashmi. What are your strengths?

Candidate: My strengths are quick learning, problem-solving, teamwork, and adaptability. I always try to understand the concept clearly rather than just memorizing it.

Interviewer: Good. Since you mentioned databases, let's move to a technical question. What does ACID stand for?

Candidate: ACID stands for Atomicity, Consistency, Isolation, and Durability.

Interviewer: What are ACID properties?

Candidate: ACID properties are rules that make database transactions safe, reliable, and accurate.

Interviewer: What is Atomicity?

Candidate: Atomicity means all operations in a transaction happen completely, or none of them happen.

Example: In a bank transfer, money should be deducted from one account and added to another. If one operation fails, the whole transaction is cancelled.

Interviewer: What is Consistency?

Candidate: Consistency means the database should always remain correct and valid after a transaction.

Example: If ₹5,000 is transferred from an account, the account balance must be updated correctly.

Interviewer: What is Isolation?

Candidate: Isolation means multiple transactions should not interfere with each other.

Example: If two people try to book the last movie ticket at the same time, only one person should successfully get the ticket.

Interviewer: What is Durability?

Candidate: Durability means that once a transaction is completed, the data is permanently saved, even if the system crashes.

Example: After a successful bank payment, the transaction should remain saved even if the server suddenly shuts down.

Interviewer: Why are ACID properties important?

Candidate: They ensure that database transactions are reliable, accurate, and secure.

Interviewer: Give a real-world application where ACID properties are critical.

Candidate: Banking systems are a good example because money transactions must be accurate and reliable.

Interviewer: Which ACID property means “all or nothing”?

Candidate: Atomicity.

Interviewer: Which ACID property ensures that committed data is not lost?

Candidate: Durability.

Interviewer: Can you explain ACID properties with one real-world example?

Candidate: Yes. Consider a bank money transfer. Atomicity ensures the complete transfer happens or none of it happens. Consistency maintains the correct account balance. Isolation prevents simultaneous transactions from interfering with each other. Durability ensures the completed transaction remains saved even after a system failure.

GeoTag:

